

THE COMPLETE ANIMATED CHART WORKFLOW

A start-to-finish system for turning an idea into researched data, a LivingCharts animation, and social-ready content.

Includes three cloneable LivingCharts templates, an AI-first CSV workflow with no R required, exact reusable prompts, design rules, image/background guidance, export QA, and publishing checklists.

Course edition: August 2026

Course map

Module	What it teaches
1	The Production System
2	Idea Selection: What Is Worth Charting?
3	Choose the Right Chart Type
4	Research and Lock the Definition
5	Build the Dataset With AI — No R Required
6	Validate the Data Before You Touch the Design
7	Images, Logos, and Entity Identity
8	The Three LivingCharts Templates
9	Importing and Mapping Data in LivingCharts
10	The Visual System: Layout, Fonts, Colors, Labels
11	Subtitle Workflow
12	Background Image Workflow
13	Animation Timing and Motion
14	Preview, QA, and Export
15	Social Packaging: Hook, Caption, Sources, Pinned Comment
16	Performance Review and Iteration
17	Master Prompt Library
18	Troubleshooting
19	Complete Production Checklists
20	Operating Principles

Core principle The course is not about memorizing software. It teaches a repeatable production system. LivingCharts is the rendering layer; the durable skill is choosing a compelling comparison, defining it correctly, structuring the data, and presenting it so a viewer understands the story in seconds.

1. The Production System

A high-performing animated chart is not one task. It is a chain of decisions. Weakness anywhere in the chain can kill the result: a boring idea cannot be rescued by polish, clean design cannot rescue incorrect data, and a strong dataset can still fail if the viewer cannot decode the chart quickly.

THE END-TO-END PRODUCTION SYSTEM



Figure 1. The production system used throughout this course.

What each stage is responsible for

Idea: Find a comparison with an obvious human reason to care. The viewer should understand the conflict, race, ranking, divergence, or surprise without reading a paragraph.

Research: Define exactly what is being measured, over what period, in what unit, and from which source. Lock definitions before collecting values.

Data: Collect, clean, transform, validate, and export a wide/transposed CSV. AI can perform this work programmatically, so students do not need to learn R for the workflow.

LivingCharts: Clone a template, import the new dataset, map identity columns, and confirm the animation behaves correctly.

Design: Apply a strict hierarchy: title, subtitle, chart, labels, imagery, background, source. Decorative elements never outrank comprehension.

Export: Watch the entire video, check key moments, render at the correct aspect ratio, and verify the final file.

Publish: Package the same underlying chart for each platform, preserve source credibility, and use performance data to choose the next idea.

The important shift Do not teach students to “write code to make a CSV.” Teach them to specify the dataset correctly and make ChatGPT produce the CSV. Coding is an implementation detail. The specification is the transferable skill.

2. Idea Selection: What Is Worth Charting?

The most expensive mistake is spending an hour producing a chart whose premise was weak from the beginning. Idea selection should be treated as a filter, not as inspiration theater.

The six-part idea score

Factor	Question	0–2 scoring rule
--------	----------	------------------

Factor	Question	0–2 scoring rule
Recognition	Will the target audience recognize the entities immediately?	0 = obscure; 1 = niche; 2 = instantly recognizable
Conflict / comparison	Is there a natural “who wins?” or “what changed?” story?	0 = none; 1 = mild; 2 = obvious
Movement	Will the visual materially change over time?	0 = flat; 1 = some movement; 2 = frequent shifts/overtakes
Surprise	Is the likely result non-obvious or counterintuitive?	0 = expected; 1 = somewhat; 2 = strong reveal
Data quality	Can the metric be sourced and defined cleanly?	0 = dubious; 1 = workable; 2 = authoritative
Visual fit	Can the story be understood in a vertical chart without explanation?	0 = poor; 1 = workable; 2 = excellent

Treat 9–12 as strong, 6–8 as conditional, and 0–5 as a warning. This is not mathematics handed down from a mountain; it is a forcing function that prevents “I found some numbers” from masquerading as an idea.

Reliable idea archetypes

- Rankings over time: companies, teams, countries, politicians, products, platforms, creators.
- Head-to-head trajectories: two to five assets, candidates, indices, brands, or outcomes.
- Cumulative races: championships, returns, wins, subscribers, revenue, debt, medals.
- Before/after or regime comparisons: presidencies, eras, rule changes, product launches, rate cycles.
- Composition shifts: market share or category share when the total is meaningful.
- “What people think vs. what the data says” comparisons, provided the underlying definitions are defensible.
- Historical reversals: an early leader loses, a small entrant overtakes incumbents, or a dominant category collapses.

Ideas to reject quickly

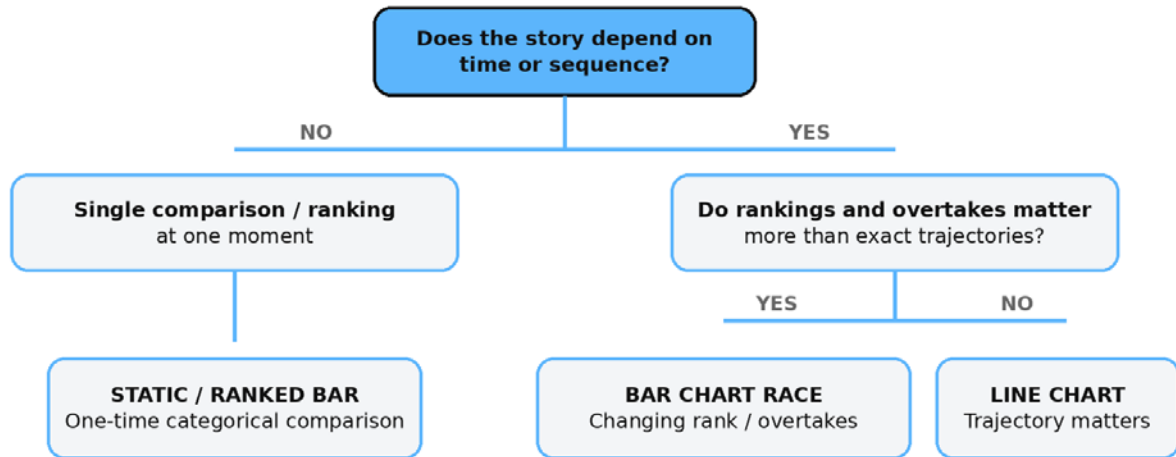
- The result is obvious before the animation starts and nothing changes.
- The metric sounds impressive but cannot be defined precisely.
- The data only exists at two or three time points, so animation adds fake drama rather than information.
- The chart needs a paragraph of caveats before a viewer can interpret it.
- The categories are too obscure for the intended audience.
- The source mixes estimates, incompatible units, or inconsistent methodologies across entities.
- The premise depends on a misleading denominator or cherry-picked start date.

Reusable idea-generation prompt

Generate 30 animated-chart ideas for [NICHE]. Prioritize ideas that have: (1) recognizable entities, (2) a clear comparison or race, (3) meaningful movement over time, (4) an interesting or non-obvious outcome, (5) data that can be sourced from authoritative public sources, and (6) strong vertical-video readability. For each idea, recommend bar chart race, line chart, ranked bar chart, or pie/share chart; specify the ideal time granularity; name likely primary sources; and score Recognition, Conflict, Movement, Surprise, Data Quality, and Visual Fit from 0–2. Sort by total score. Do not invent values.

3. Choose the Right Chart Type

WHICH CHART TYPE SHOULD YOU USE?



PIE / DONUT: Use only when categories are shares of one meaningful total and there are few slices.

Avoid forcing every idea into a race. Motion is a storytelling device, not a tax requirement.

Figure 2. Use motion only when sequence or changing rank is part of the story.

Bar chart race

Use a bar chart race when the central story is changing rank among many categories. It is strongest when viewers care about leaders, overtakes, entrants, collapses, and the order of the field. LivingCharts specifically supports bar chart races and describes them as a way to show how categorical values and rankings change over time (see the official LivingCharts site and bar-chart-race tutorial in the references section).

- Best with roughly 5–15 visible categories at a time.
- Use consistent categories and time intervals.
- Choose enough time points for smooth motion rather than abrupt yearly jumps when monthly or quarterly data exists.
- Cumulative metrics work especially well because the viewer can track durable rank changes.

Line chart / line race

Use a line chart when the path matters more than rank. A line lets the viewer see acceleration, drawdowns, divergence, convergence, and turning points. Keep the number of series small enough that a phone-sized viewer can follow them.

Pie / share animation

Use a pie or share chart only when every slice is part of one meaningful whole. Market share can qualify. Unrelated totals do not. If the values do not sum to a coherent total, a pie chart is mathematically decorative nonsense with rounded corners.

Static or ranked bar chart

Use a static/ranked bar when time is not necessary. If the story is simply “who is largest right now,” animation may weaken the information density rather than improve it.

4. Research and Lock the Definition

Before collecting numbers, write a one-sentence metric definition. This prevents the most common data error: starting with one concept and quietly switching to a different one because the second source was easier to find.

Definition lock Write: “This chart measures [METRIC], using [UNIT / ADJUSTMENT], for [ENTITIES], from [START] through [END], at [FREQUENCY], sourced primarily from [SOURCE].” If this sentence is fuzzy, the dataset is not ready.

Source hierarchy

Tier	Prefer	Examples
1	Primary / official	Government databases, company filings, league/federation records, official platform APIs, central banks
2	High-quality aggregators	Established datasets that clearly document methodology and provenance
3	Secondary reporting	Reputable journalism or research summaries used to corroborate, not replace, a better primary source
4	Discovery only	Search results, social posts, blogs, unsourced graphics. Useful for finding leads; poor as final evidence

Questions to settle before data collection

- Nominal or inflation-adjusted?
- Price level, percent change, cumulative change, or annualized rate?
- Calendar year, fiscal year, season, month, quarter, trading day, or event number?
- End-of-period value, average value, median, total, or point estimate?
- Are entities being compared on the same definition?
- What happens when data is missing?
- Are revisions possible? If so, what “as of” date is being used?
- Is the intended conclusion sensitive to the start date?

Never mix definitions silently

Revenue is not profit. Market capitalization is not revenue. Average inflation during a term is not total inflation over the term. Nominal dollars are not real dollars. Votes cast are not eligible voters. A chart can be visually perfect and still be garbage if the denominator changes halfway through.

Research specification prompt

I am building an animated chart about [TOPIC]. Before collecting values, define the metric precisely. State the unit, date range, frequency, entities, preferred primary source, fallback source, treatment of missing values, and every transformation required. Identify likely methodological traps or incompatible definitions. Do not collect the final data until the definition is internally consistent.

5. Build the Dataset With AI — No R Required

The student does not need R. ChatGPT can research the values, perform the transformations in Python or another internal computation layer, validate them, and return the final CSV file. The student's job is to specify the transformation correctly and review the result.

LIVINGCHARTS DATA SHAPE: WIDE / TRANSPOSED

Time goes across columns. Entities go down rows. Numeric cells contain the values.

Name	Image URL	Jan 2024	Feb 2024	Mar 2024	Apr 2024
Alpha	https://.../a.png	92	101	117	125
Beta	https://.../b.png	110	106	112	119
Gamma	https://.../c.png	75	88	98	121

Avoid: Currency symbols inside numeric cells; merged cells; inconsistent date labels; mixed units; or prose in value columns.

Figure 3. The preferred wide/transposed format for animated time-series charts.

The required data orientation

For the common LivingCharts workflow used in this course, categories/entities are rows and time points are columns. Optional identity columns such as image URL can sit beside the entity name. LivingCharts' official tutorial says data can be entered manually, pasted, or imported from files including CSV, ODS, XLS, and XLSX, and its data settings can map title, image, and category columns.

What ChatGPT should return

- A downloadable .csv file, not merely a code snippet.
- One row per entity.
- One column per time point in chronological order.
- Pure numeric values inside value cells unless the chart explicitly requires text.
- Optional Image URL and Category columns when needed.
- A short data dictionary describing every field.
- A source ledger listing which source supports which portion of the dataset.
- A validation note covering missing values, transformations, and any estimates.

Master CSV-builder prompt

Research and build the complete dataset for this animated chart: [DESCRIBE CHART]. Use [DATE RANGE] at [MONTHLY/QUARTERLY/ANNUAL] frequency. Use authoritative primary sources whenever possible. Lock the metric definition before collecting values and do not mix incompatible definitions. Perform any required calculations yourself. Return a downloadable CSV in wide/transposed format: first column = entity name; second column = Image URL if applicable; remaining columns = chronological time periods; one entity per row. Numeric cells must contain only numbers. Also return a source ledger, transformation notes, and validation checks. Do not give me R code. Generate the CSV file directly.

Common transformations AI can do for you

Transformation	When to use it	Instruction language
Index to 100	Compare growth paths with different starting magnitudes	“Normalize every series to 100 at the first valid observation.”
Percent return	Asset/performance comparison	“Calculate cumulative percent return from the common start date.”
Month 1 → Month N	Compare terms, presidencies, seasons, product lifecycles	“Align each entity by elapsed month rather than calendar date.”
Inflation adjustment	Compare historical dollar amounts in real terms	“Convert all nominal values to [BASE DATE] dollars using [INDEX].”
Cumulative total	Championships, wins, debt added, units sold	“Convert event-level values into a running cumulative total.”
Share of total	Market-share style composition	“At each time point, divide each entity by the total across included entities.”
Rolling average	Reduce noisy short-term movement	“Compute a trailing N-period average and preserve the original dates.”

Granularity matters

More time points usually produce smoother motion, but “more” is not automatically “better.” Use the highest meaningful and reliable frequency. Daily data can create jitter; annual data can make a 20-second animation look like a slideshow. Monthly or quarterly data is often a strong compromise for multi-year stories.

Missing data rules

- Do not convert “unknown” into zero unless zero is truly the measured value.
- If an entity did not yet exist, decide whether blank, zero, or later entry is conceptually correct.
- Interpolation is acceptable only when the metric and course methodology explicitly justify it; label estimates as estimates.
- If a source changes methodology, either harmonize the series or split/flag the break.

6. Validate the Data Before You Touch the Design

Do not design around unverified numbers. Once a chart is styled, humans become emotionally attached to it and mysteriously less interested in discovering that column 38 was off by a factor of 1,000.

Minimum validation pass

- Check the first time point for every entity.
- Check the final time point for every entity.
- Check every dramatic jump, crossover, spike, or collapse.
- Confirm units and scaling.
- Confirm dates are chronological and consistently formatted.
- Confirm no duplicate entities or duplicate dates.
- Confirm calculations against at least one hand-worked example.
- Confirm the chart title describes exactly what the numbers represent.
- Preserve sources before moving into design.

AI validation prompt

Audit this dataset before I import it into LivingCharts. Check: chronological order, duplicate dates, duplicate entities, nonnumeric value cells, missing-value handling, unit consistency, implausible jumps, arithmetic transformations, start/end values, and whether the title/subtitle accurately describe the metric. For every suspicious point, identify the row, column, value, and reason. Do not silently “fix” anything without explaining the change.

7. Images, Logos, and Entity Identity

Entity images can make a moving chart dramatically easier to decode because the viewer recognizes a face, team logo, flag, or brand mark before reading the label. They can also turn a clean chart into a sticker collection assembled by a caffeinated raccoon. Use them selectively.

Use images when

- The entity is visually recognizable: person, team, company, country, product, platform.
- The image reduces reading time.
- The number of visible entities is limited enough that images remain distinct.
- The image source is stable and you have the right to use it.
- The logo/headshot remains legible at phone scale.

Skip images when

- The chart already contains many categories and the images create clutter.
- The identity is abstract or the image adds no recognition advantage.
- The available images have wildly inconsistent crops or backgrounds.
- The image URL is unstable or blocks hotlinking.
- The image creates contrast problems against the assigned bar/line color.

Image URL workflow

1. Ask ChatGPT for one direct image URL per entity, preserving the exact row order of the CSV.

2. Prefer stable, reputable hosts and consistently cropped images.
3. Add the URLs to an Image URL column rather than manually attaching images one by one.
4. Import the updated CSV and map that column as the image field in LivingCharts. The official tutorial documents image-column mapping in the data settings panel.
5. Preview several points in the animation. A URL that works in your browser can still fail inside another renderer.

Image URL prompt

For the following entities, return one stable direct image URL per entity in exactly this order: [LIST]. Prefer official, Wikimedia Commons, league/team, company, or other durable sources. The URL must point directly to an image file when possible. Use consistent headshot/logo style across the set. Return a two-column CSV: Name, Image URL. Do not reorder the entities.

Color selection when an image sits on top of a bar

Do not blindly use the official brand color if the logo disappears into it. The purpose of color is identity plus legibility. If a white logo sits on a very light brand color, darken the bar. If a dark logo sits on a near-black bar, lighten the bar. Recognition wins over brand-palette purity.

8. The Three LivingCharts Templates

These three project links are the course's canonical starting templates. Students should clone a template rather than rebuilding the design from scratch. This preserves the layout, typography, spacing, animation behavior, and other project settings already established in the template.

CLONE FIRST. EDIT THE COPY.

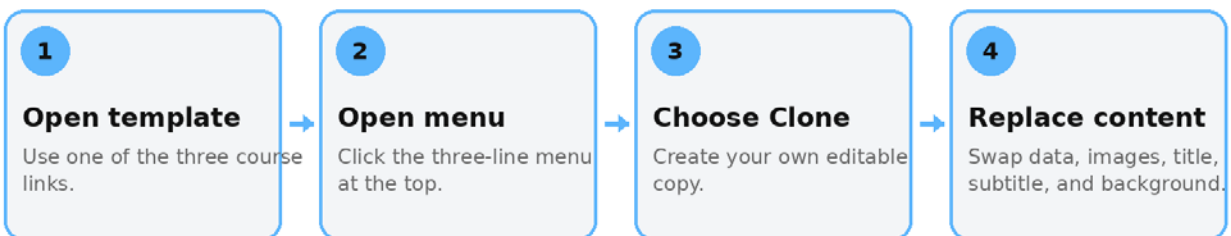


Figure 4. The template workflow. Never edit the shared original when a clone is available.

Exact clone instruction Open the template. Click the three-line menu at the top of the page. Click “Clone.” Work from the cloned copy. Replace the data and creative assets inside your own project.

**Template 1**[Open template](https://livingcharts.com/app/edit/A4gdsZsc5gOKtLf3oFcs)

<https://livingcharts.com/app/edit/A4gdsZsc5gOKtLf3oFcs>

**Template 2**[Open template](https://livingcharts.com/app/edit/X76zY6Wh4mG93g1SlaDb)

<https://livingcharts.com/app/edit/X76zY6Wh4mG93g1SlaDb>

**Template 3**[Open template](https://livingcharts.com/app/edit/luv8JESBP3dxBVPvYYVO)

<https://livingcharts.com/app/edit/luv8JESBP3dxBVPvYYVO>

Because LivingCharts' editing interface is JavaScript-rendered, its public crawler does not expose the internal configuration of these specific project IDs. Treat the cloned project itself as the source of truth for its exact settings. The stable course rule is therefore: clone first, then replace content without resetting the entire project.

What you normally replace after cloning

- Dataset / CSV
- Title
- Subtitle asset
- Entity images or logos
- Entity colors
- Background image
- Units, prefixes, suffixes, and date labels
- Animation duration if the dataset length materially differs
- Source line / attribution
- Any project-specific annotations

What you normally preserve until there is a reason to change it

- Aspect ratio
- Overall layout
- Text hierarchy
- Default font family
- Element positions
- General spacing
- Animation behavior that already works in the template

9. Importing and Mapping Data in LivingCharts

LivingCharts' official tutorial describes two core editor views: Data and Preview. Data is where the table and field mappings live; Preview is where the animation and project settings are adjusted. Data can be entered manually, pasted with Ctrl+V, or imported from supported spreadsheet files.

Recommended import sequence

1. Clone the correct course template.
2. Open the Data view.

3. Import the final CSV rather than partially editing the old dataset in place.
4. Verify the entity-name column.
5. Map Image URL if the chart uses images.
6. Map category/group fields if the template uses them.
7. Confirm the time columns appear in chronological order.
8. Switch to Preview and scrub through the full animation.

Immediate post-import checks

- Does every entity appear?
- Do rankings/order match the source data at the start?
- Does the date label display correctly?
- Do values have the right prefix/suffix?
- Are images mapped to the correct names?
- Do colors remain attached to the intended entities?
- Do missing values behave as intended?
- Is the chart accidentally reading an Image URL or Category column as numeric data?

LivingCharts supports customization of animation duration, FPS, aspect ratio, background, label/order presentation, bar colors, value prefixes/suffixes, and additional elements such as text, graphs, or images.

10. The Visual System: Layout, Fonts, Colors, Labels

The visual goal is not “make it pretty.” The goal is to minimize decoding time while preserving enough identity that the content feels native to the topic.

VERTICAL VISUAL HIERARCHY

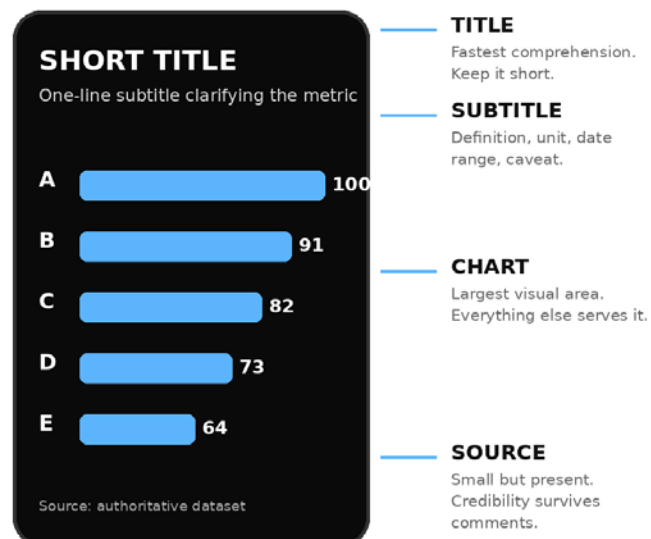


Figure 5. A simple vertical hierarchy: title → subtitle → chart → source.

Course baseline preset

Element	Starting point	Why
Aspect ratio	9:16 vertical	Native framing for Reels, TikTok, and Shorts
Background	Near-black / black	Maximum contrast and lets imagery sit underneath at low opacity
Primary accent	#5CB5FC	Readable cool-blue accent that works against black
Font	Roboto Condensed or a similarly narrow sans-serif	High information density without shrinking type excessively
Title	Short, bold, one or two lines maximum	The hook must be readable before the animation is understood
Subtitle	Usually one line; two lines only when necessary	Clarifies metric without becoming body copy
Source	Small and unobtrusive but readable	Credibility without stealing hierarchy
Background image opacity	Start near 0.15; usually test ~0.12–0.20	Context remains visible without fighting the chart

These are defaults, not laws. The template should do most of this work. Change a default only because the topic or legibility requires it, not because sliders exist and human beings apparently feel obligated to touch all of them.

Titles

- Lead with the comparison, not the methodology.
- Prefer concrete nouns and familiar entities.
- Remove filler words until the title still makes sense but cannot be shortened without losing meaning.
- Do not force the unit into the title if the subtitle can carry it more cleanly.
- Avoid announcing the chart type (“Bar Chart of...”). The viewer has eyes.

Labels and values

- Choose a prefix/suffix that prevents repeated explanatory text: \$, %, B, M, points, votes, etc.
- Use consistent decimal precision across entities.
- Do not show two decimals when the source is only precise to whole units.
- Shorten large numbers only when the abbreviation is obvious to the audience.
- Ensure label color contrasts with both bars and background.

Entity colors

For teams, companies, political parties, or products with established identity colors, start from those colors. Then modify lightness if necessary to maintain contrast with the entity image and text. For charts without natural identity colors, use a restrained palette rather than assigning a different neon color to every row.

11. Subtitle Workflow

The subtitle is where precision lives. The title earns attention; the subtitle tells the viewer what the chart actually measures.

A strong subtitle answers one or more of these

- What exactly is being measured?
- What unit is used?
- What date range or alignment is used?
- Is the value cumulative, annualized, indexed, inflation-adjusted, or estimated?
- What important caveat would otherwise make the title misleading?

One line vs. two lines

Default to one line. Use two only when shortening further would remove essential meaning. If two lines are required, split at a natural grammatical boundary and keep the lines visually balanced.

Generate the subtitle text first

Subtitle-copy prompt

Write the shortest accurate subtitle for this chart: [TITLE + METRIC DESCRIPTION]. It must clarify the metric, unit, date range/alignment, and any essential adjustment without repeating the title. Prefer one line. Give me one recommended version and two shorter alternatives. No hype.

Generate the subtitle image

If the template uses the subtitle as an image asset, have ChatGPT generate the PNG programmatically so the text is exact. This is preferable to generative-image text rendering, which remains an unnecessarily creative way to misspell a five-word sentence.

Exact subtitle-PNG prompt

Create a PNG subtitle asset with the exact text below. Do not use generative image synthesis; render the text programmatically so every character is exact. Canvas: [WIDTH] × [HEIGHT] px. Background: transparent unless I specify black. Font: Roboto Condensed (use the closest available narrow sans-serif if unavailable). Text: white. Alignment: left. Fit the text on one line if possible; if I specify a line break, preserve it exactly. Tight crop with minimal padding. Return the PNG as a downloadable file. Text: “[SUBTITLE]”

Subtitle QA

- Read it at phone size, not on a 27-inch monitor.
- Confirm every character and number.
- Check that it does not duplicate the title.
- Check that its definition exactly matches the dataset.
- Confirm the crop has no excessive empty pixels that make placement annoying.

12. Background Image Workflow

A background image gives the chart topic-specific atmosphere. It should be recognizable only after the viewer has already understood the title and chart. If the background is the first thing you notice, it is too strong.

BACKGROUND IMAGES SHOULD SUPPORT THE CHART, NOT COMPETE WITH IT

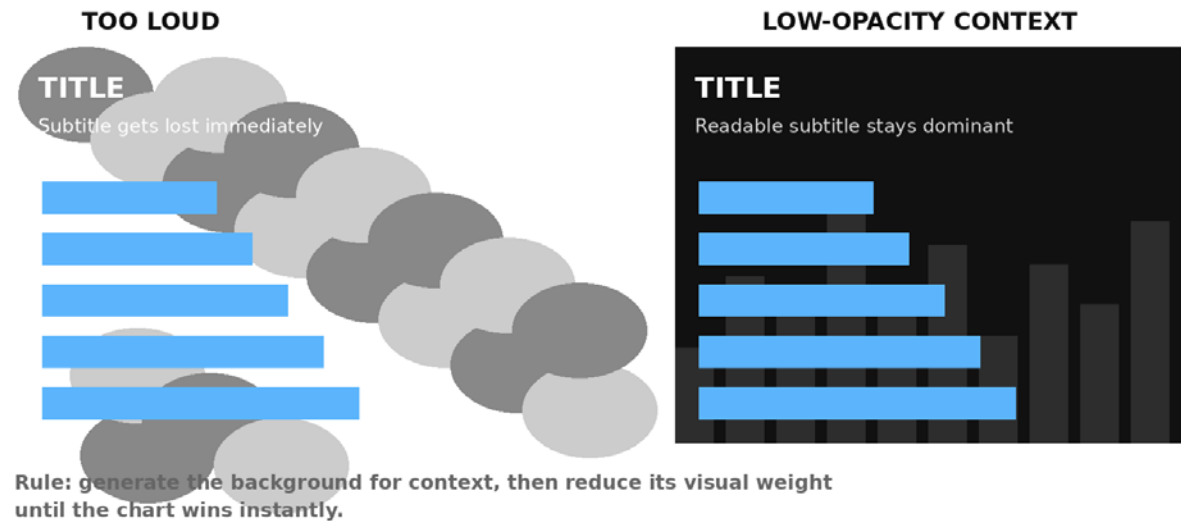


Figure 6. Background imagery should survive as context, not as competition.

When a background helps

- The subject has a recognizable environment: trading floor, Capitol, stadium, city skyline, map texture, product category, historical scene.
- The background can communicate the niche instantly without additional text.
- The central area can remain uncluttered enough for chart labels.
- The image can be darkened substantially and still retain useful context.

When to skip it

- The chart is already image-heavy.
- The background requires bright faces, logos, text, or fine detail to make sense.
- The subject does not have an obvious visual environment.
- The background causes low contrast or makes labels appear to vibrate during motion.

How to generate the background

9:16 background-generation prompt

Create a 9:16 vertical background image for an animated data visualization about [TOPIC]. It will sit behind moving chart elements at low opacity, so optimize for silhouette, atmosphere, and large shapes rather than small details. Keep the center and upper text zones relatively uncluttered. No readable text, numbers, labels, watermarks, UI, or chart elements. Dark cinematic lighting, realistic rather than obviously AI-generated, with enough contrast to remain visible around 15% opacity. The chart itself will be overlaid later.

Opacity process

1. Start around 0.15 opacity.
2. Preview at normal phone size.
3. Lower it if it competes with labels, bars, faces, or values.
4. Raise it slightly only if the topic is no longer visually identifiable.
5. Test roughly 0.12–0.20 rather than assuming one number works for every image.

Composition rule Generate the background for the chart layout. Do not generate a beautiful standalone poster and then wonder why there is nowhere to put the data.

13. Animation Timing and Motion

Animation speed controls comprehension. Too slow and the viewer leaves before the story develops. Too fast and overtakes happen before the viewer can register them.

Starting point

For short-form social content, a roughly 20–30 second animation is a useful starting range; around 23 seconds is a practical default for the course templates. Adjust based on the number of time points and the density of rank changes. This is a production heuristic, not a platform rule.

How to judge timing

- Watch the first three seconds. Is there immediate movement or a clear starting state?
- Watch the most important crossover. Can you see it happen without scrubbing?
- Watch the date label. Can you mentally follow the timeline?
- Watch the bottom-ranked visible categories. Do they flicker in and out too quickly?
- Watch the final two seconds. Does the result remain visible long enough to register?

Duration should scale with information

A four-series line chart can move faster than a 15-category bar race. A 50-year annual series may need less time than a three-year monthly series if the monthly version contains much more visible movement. Optimize for events per second, not years per second.

14. Preview, QA, and Export

LivingCharts’ official documentation supports direct animation export to video and image sequences, and its tutorial notes that rendering occurs in the browser and may slow in some browsers if the tab is inactive.

Full preview checklist

- Watch from first frame to last without scrubbing.
- Check spelling in title, subtitle, labels, and source line.
- Check every entity image at least once.
- Check the most extreme numeric value.
- Check the biggest jump and most important crossover.

- Check that no label leaves the safe area.
- Check the chart at phone size.
- Check the final state against the source dataset.
- Check that background opacity remains subtle throughout.
- Check that the title/subtitle do not collide with chart elements.

Export baseline

- Use the cloned vertical template/aspect ratio for short-form platforms.
- Export to video at the highest practical vertical resolution offered by the project/template.
- Keep the rendering tab active if your browser slows background rendering.
- Open the rendered file after export and watch it again. Preview is not proof that the exported file is clean.

Export rule Never publish directly from the moment the render completes. Watch the actual exported file once. This catches missing images, dropped frames, incorrect crop, bad text, and other tiny acts of software sabotage.

15. Social Packaging: Hook, Caption, Sources, Pinned Comment

The video is the product; the caption is packaging. Its job is to frame the comparison, invite interpretation without lying about the data, and preserve source credibility.

Caption structure

1. Opening line: state the comparison or surprising pattern plainly.
2. One or two sentences: explain what the viewer is looking at or note the most important methodological detail.
3. Optional engagement line: ask for interpretation only when there is a genuine comparison people can debate.
4. Follow CTA: short and topic-specific.
5. Sources: list the authoritative sources used for the dataset.

Do not do this

- Claim causation when the chart only shows correlation or timing.
- Hide a major methodological caveat because it weakens the hook.
- Use “data proves” when the metric is an estimate or one possible definition.
- Write a six-paragraph caption for a 23-second chart unless the platform/audience genuinely rewards it.
- List “Google” as a source. Search is how you find a source; it is not the source.

Caption prompt

Write the social caption for this chart. Title: [TITLE]. Subtitle/definition: [SUBTITLE]. Key pattern: [WHAT HAPPENS]. Sources: [SOURCES]. Requirements: concise opening hook, accurate explanation, no unsupported causal claims, one natural engagement line if the topic supports debate, a tailored follow CTA, then sources at the end. Preserve political/financial neutrality unless the chart itself is explicitly opinion content.

Pinned comment

Use the pinned comment for a second interaction surface: clarify a surprising result, ask a narrow comparison question, or reinforce the follow CTA. Do not simply paste the caption again.

16. Performance Review and Iteration

Treat each chart as a test of several variables at once: topic, hook, chart type, duration, visual identity, background, title, and distribution. The goal is not to explain every view count perfectly. The goal is to identify repeatable patterns.

Track at minimum

- Topic / niche
- Chart type
- Post date/time
- Video length
- Views
- Average watch time or retention when available
- Shares
- Saves
- Comments
- Follows attributed to the post when available
- Whether the post was a remake/variation of a previous winner

Winner adjacency

When a chart wins, do not immediately abandon the concept to prove your creativity. Create adjacent versions that preserve the underlying reason people cared while changing one major variable: entity set, date range, metric, category, framing, or chart type.

If the winner was...	Test next...
A presidency comparison	Another household/economic metric using the same alignment
A company ranking race	Revenue, market cap, profit, users, or units sold using the same entity set
A two-asset line comparison	Add one meaningful competitor or change the start regime
A cumulative sports race	Different league/era/achievement using the same structure
A market-share animation	A longer date range or category-level decomposition

Diagnose failure in order

1. Was the idea intrinsically interesting to the target audience?
2. Was the title understood instantly?
3. Did the first frame look worth watching?
4. Did the animation produce movement soon enough?
5. Was the chart readable on a phone?
6. Did the result actually contain a reveal?

7. Was the data trustworthy enough that comments did not derail the post?
8. Only then worry about microscopic visual tweaks.

17. Master Prompt Library

The prompts below are deliberately explicit. Vague prompts produce vague data, then someone spends an hour fixing a spreadsheet that never needed to be wrong.

A. Full idea-to-data prompt

I want to create a viral but accurate animated chart for [AUDIENCE/NICHE]. Topic direction: [DIRECTION]. First propose 15 ideas and score them for recognition, conflict, movement, surprise, data quality, and visual fit. For the top 3, recommend chart type and data frequency, identify primary sources, and explain the exact metric definition. Do not invent values. Stop after the plan.

B. Full research + CSV prompt

Build the final dataset for: [CHART]. Definition: [DEFINITION]. Date range: [RANGE]. Frequency: [FREQUENCY]. Entities: [ENTITIES]. Use primary sources whenever possible and document all sources. Perform the calculations yourself. Return a downloadable wide/transposed CSV with entity names in rows and dates in chronological columns. Add Image URL and Category columns only if needed. Numeric value cells must contain only numbers. Also return a validation report and source ledger. No R code.

C. Month-aligned comparison prompt

For each entity/term in [LIST], align the series by elapsed month: Month 1, Month 2, ... Month N. Use the exact start date for each entity. Define whether Month 1 means first full month or the start-date month, and apply that rule consistently. Return a wide/transposed CSV with one row per entity and Month 1...Month N across columns. Document the original calendar dates used for each aligned month.

D. Inflation-adjustment prompt

Convert each historical nominal value to [BASE MONTH/YEAR] dollars using [CPI/INDEX]. State the formula, identify the exact index series and base period, calculate the adjusted values programmatically, and verify at least three points by hand. Return the adjusted wide/transposed CSV and a second audit table showing nominal value, index at observation, base index, and adjusted value for the checked points.

E. Color prompt

Assign one bar/line color hex to each entity in exactly this order: [LIST]. Start from recognizable official/brand/team/party colors where appropriate, but optimize final colors for readability on a near-black background and for contrast with the entity logo/headshot placed over the color. Return Name, Hex, and a one-sentence contrast note. Do not reorder.

F. Title + subtitle prompt

For this chart concept and definition: [PASTE], produce 10 short titles optimized for immediate comprehension, not clickbait ambiguity. Then produce one recommended one-line subtitle that states the metric/unit/date alignment without repeating the title. Rank the titles by clarity and curiosity.

G. Background prompt

Create a 9:16 background image for [TOPIC]. It will be used behind an animated chart at roughly 15% opacity. Use large recognizable shapes, dark realistic lighting, uncluttered center/top areas, and no readable text, watermarks, charts, labels, or UI. It should communicate the topic after being heavily subdued, not function as the main image.

H. Design critique prompt

I am attaching a screenshot of my chart. Critique it specifically for vertical-phone viewing. Evaluate title hierarchy, subtitle readability, label contrast, image/logo legibility, background opacity, bar/line colors, spacing, safe zones, and whether the chart can be understood in under two seconds. Give exact changes, not general design advice.

I. Final QA prompt

Audit the final chart package before publication. Compare the exported video/screenshot against the source dataset and this definition: [DEFINITION]. Check factual accuracy, title/subtitle alignment, units, dates, images, colors, source line, potential misleading interpretations, readability, and any element that may fail on a phone screen. Return PASS/FAIL for each category and exact corrections for every failure.

18. Troubleshooting

Problem	Likely cause	Fix
Chart imports incorrectly	CSV orientation or mapping is wrong	Confirm entities are rows, time is columns, and identity columns are mapped rather than treated as values.
Values show as text / do not animate	Symbols or commas are embedded inconsistently	Keep numeric cells numeric; use chart prefix/suffix settings for \$, %, M, B, etc.
Dates appear out of order	Inconsistent or ambiguous labels	Use one date format and chronological columns; verify before import.
Entity images mismatch	Rows reordered between data and image URL generation	Force exact order; merge by entity name rather than visual copy/paste.
Logos disappear	Bar color and logo share similar lightness	Darken/lighten the bar color while preserving identity.
Background looks AI / distracting	Too much detail, text, faces, or contrast	Regenerate for low-opacity use; simplify composition; reduce opacity.

Problem	Likely cause	Fix
Subtitle looks blurry	Raster asset too small or scaled excessively	Generate at final or larger pixel dimensions and crop tightly.
Animation feels dead	Too few time points or metric barely moves	Increase meaningful frequency or reject the idea.
Animation feels chaotic	Too many categories / too much short-term noise	Reduce visible categories, aggregate frequency, or use line/static format.
Render is slow	Browser rendering and complex assets	Keep the tab active; simplify excessive imagery if needed. LivingCharts notes browser rendering can slow when inactive.
Export looks different from preview	Rendering/crop/image-load issue	Watch the exported file; re-render after correcting the asset or project setting.
Comments attack methodology	Definition/caveat was not explicit	Strengthen subtitle/source note; verify the metric before reposting.

When to rebuild instead of patch

Rebuild the dataset when the metric definition changes. Do not patch a few columns and pretend the rest of the series is still comparable. Rebuild the design only when the template fundamentally conflicts with the story. Most visual problems are one-setting problems; most methodological problems are whole-dataset problems.

19. Complete Production Checklists

A. Before research

- ☐ Audience is defined.
- ☐ Idea scores strongly on recognition/conflict/movement/surprise/data quality/visual fit.
- ☐ Chart type is justified by the story.
- ☐ Metric definition is written in one sentence.
- ☐ Start/end dates and frequency are fixed.

B. Before CSV export

- ☐ Primary sources identified.
- ☐ Definitions are consistent across entities.
- ☐ Transformations documented.
- ☐ Missing values handled explicitly.
- ☐ First, last, and dramatic points verified.
- ☐ Dates are chronological.
- ☐ Numeric cells are numeric only.
- ☐ Image URLs/colors preserve entity order.

C. Before LivingCharts export

- ☐ Correct template cloned.
- ☐ Data imported and mapped.
- ☐ Title is short.

- ☐ Subtitle is accurate.
- ☐ Images/logos are legible.
- ☐ Colors have sufficient contrast.
- ☐ Background is contextual and subdued.
- ☐ Animation speed is readable.
- ☐ Source line is present.
- ☐ Full animation previewed.

D. Before publishing

- ☐ Exported file watched in full.
- ☐ Final state matches dataset.
- ☐ No spelling errors.
- ☐ Caption does not overclaim.
- ☐ Sources are included.
- ☐ Pinned comment adds a useful interaction point.
- ☐ Post is logged for later performance comparison.

20. Operating Principles

Idea > execution. A perfectly rendered boring comparison is still boring.

Definition before data. The metric sentence comes before the spreadsheet.

Data before design. Do not polish a dataset you have not validated.

Clone before rebuilding. The templates exist to remove repetitive layout work.

Recognition before decoration. Images and colors are useful when they reduce decoding time.

Subtitle = precision. Put the methodology needed for interpretation there, not in a hidden comment reply.

Background = context. If it becomes content, reduce it.

Motion must earn its place. Use animation to reveal change, not because the software can animate.

Export is not QA. Watch the exported file.

Sources are part of the product. A chart that cannot survive “source?” is unfinished.

Repeat winners intelligently. Preserve the reason a chart worked and change one major variable at a time.

AI replaces mechanical work, not judgment. Let AI collect, calculate, format, and export. You still own the definition, framing, validation, and visual decision-making.

The workflow in one sentence Choose a comparison people care about, define it precisely, make AI build and validate the dataset, clone the correct LivingCharts template, replace only the content that needs changing, make the chart instantly readable on a phone, export, verify, publish with sources, and use the result to choose the next chart.

One-page quick reference

Stage	Do this	Do not do this
Idea	Score the premise before researching	Spend an hour on a technically possible but emotionally inert metric
Definition	Lock unit, period, frequency, entity set, and source	Change definitions mid-series
Data	Ask AI for the finished wide CSV	Teach R unless coding itself is part of the product
Images	Use them when recognition improves	Add images merely because there is an image field
Template	Three-line menu → Clone → edit copy	Rebuild the whole project every time
Subtitle	Clarify metric and caveat	Repeat the title
Background	9:16, topic-relevant, low-detail, ~0.15 starting opacity	Use a poster-like image that competes with the chart
Timing	Make changes readable at phone size	Optimize only for shortest possible runtime
Export	Watch the rendered file	Assume preview = final
Publish	Accurate hook + source line + focused CTA	Overclaim causation or hide methodology
Iterate	Build adjacent ideas from winners	Change every variable at once

Platform references and notes

LivingCharts features and editor workflow described in this course were checked against LivingCharts' public site and official tutorials in August 2026. The three project-template URLs were supplied for this course and should be treated as the canonical source for their exact project settings.

LivingCharts homepage: <https://livingcharts.com/>

Official tutorial: How to make animated Bar Chart Race: <https://livingcharts.com/tutorials/how-to-make-bar-chart-race>

LivingCharts guide: How to Make Ranking Videos for YouTube Without Coding:
<https://livingcharts.com/blog/how-to-make-ranking-videos>

LivingCharts pricing / feature overview: <https://livingcharts.com/pricing>

Copyright and data note: verify image/logo usage rights, dataset licensing, attribution requirements, and platform rules for the material you publish. Public availability does not automatically make every asset unrestricted.